GATE

Computer Science & Information Technology

C Programming

DPP: 1

Control Flow Statements

Q1
```
#include <stdio.h>
    int main(void)
{
        int i = 2, j = 3, k = 4;
        if (i < j ? 1 : 0)
            printf("GATE");
        else
            printf("Wallah2023");
        return 0;
    }
```
The output of the program is _____.

Q2
```
#include <stdio.h>
    void main( )
   {
    int a, b, c, d;
      a = 2; b = –1; c = 3; d = –4;
      if(a = b – c – d)
          printf("%d%d%d", a++, b--, c++);
      else
          printf("%d%d%d", c--, ++ a, ++b);
    }
```
The output is _____.

(A) 1 −2 4
(B) 3 1 0
(C) 2 1 −3
(D) 3 3 0

Q3
```
#include <stdio.h>
    int main(void)
     {
     int a = 3 > 2 ? 0 ? 0 : 1 : 5;
      if(a = = a – 1)
          printf("GATE 2023");
      else
          printf("GATE WALLAH");
      return 0;
     }
```
The output of the program is _____.

(A) GATE 2023 (B) GATE WALLAH
(C) Compiler error (D) Garbage value

Q4
```
#include <stdio.h>
    void main( )
    {
        int a;
          a = printf("GATE Wallah 2023");
       if(a%4 = = 0)
          a = a + 5;
        else
          a = a – 5;
        printf("%d", a++);
    }
```
The value of a at the end of the program is ___.

Q5
```
#include <stdio.h>
    void main( )
    {
        int i, j, k;
          j = 4;
          k = 0;
        i = j < k ? k : j --;
        if (j < i)
            j = j + k – 1;
        if (j = = i)
            j = j – i;
        else
            j = j + -- k,
        printf ("%d", j + k – i);
    }
```
The output is ___.

Q6 Consider the following program:
```
#include<stdio.h>
int main()
```

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```
{
int a=19, b=20;
if(a++<b--) printf("%d",a+++--b);
else printf("%d", ++a+--b);
return 0;
}
```

The output is _____.

Q7

```
#include<stdio.h>
void main()
    {
    int a=0;
    printf("%d", a);
    if(a=2){
       printf("Hi");
       printf("%d",a);
    }else{
       printf("Bye");
    }
    printf("%d", a);
    }
```

The output string is:

(A) 0Hi22 (B) 0Hi20

(C) 0Bye0 (D) 0Hi00

Q8

```
#include<stdio.h>
void main()
    {
       int a=0, b=0;
       a=(a=4)||(b=1);
       if(a&&b) printf("CProgramming");
       else printf("PhysicsWalah");
       printf("%d",b);
    }
```

The output is-

(A) CProgramming0 (B) CProgramming1

(C) PhysicsWalah0 (D) PhysicsWalah1

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Answer Key

Q1	GATE	Q5	-4~-4
Q2	(B)	Q6	38~38
Q3	(B)	Q7	(A)
Q4	22~22	Q8	(C)

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Hints & Solutions

Q1 Text Solution:

i < j ? 1 : 0

In the above expression i value is less than j value, hence it will return 1.

So, it will print GATE.

Q2 Text Solution:

a = – 1 – 3 + 4

a = 0

$$\begin{array}{r|l} a & \cancel{2}\ \ 0 \\ \hline \end{array}$$

Assignment operator assigns and returns the value

$$\begin{array}{r|l} b & ++b: \quad \cancel{-1}\ \ 0 \\ \hline \end{array}\ ,$$

$$\begin{array}{r|l} a & ++a: \ \cancel{0}\ \ 1 \\ \hline \end{array}\ ,$$

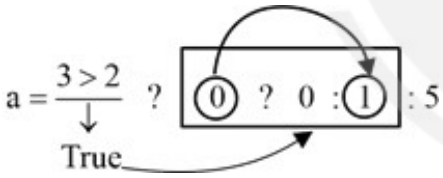
$$\begin{array}{r|l} c & c--: \ \cancel{3}\ \ 2 \\ \hline \end{array}$$

⇓

Post decrement (It will print 3, then update to 2)

Output: 3 1 0.

Q3 Text Solution:



a = 1

Assignment operator assigns the value and returns it.

$$\text{if} \quad \underbrace{(a = a - 1)}_{0}$$

⇓

Condition: false

Output: GATE WALLAH

Q4 Text Solution:

$$\text{GATE Wallah } 2023$$

$$\downarrow$$

$$a = 16$$

printf returns the number of characters successfully printed

$$16 \ \% \ 4 = 0 \rightarrow \text{True}$$

$$\downarrow$$

$$a = a + 5$$

$$\begin{array}{r|l} a & \cancel{16}\ \ \cancel{21}\ \ 22 \\ \hline \end{array}$$

Hence the final value of a is 22.

Q5 Text Solution:

$$\begin{array}{cccc} \overset{i}{4} & \overset{j}{\cancel{4}} & \overset{k}{\cancel{0}} & -1 \\ & \cancel{3} & \cancel{2} & 1 \end{array}$$

3 < 4

j = 3 + 0 – 1 = 2

2 ! = 4

j = j – 1

= 2 – 1

j = 1

printed value = j + k – i

= 1 – 1 – 4

= –4

Q6 Text Solution:

If(19<20)⟶ Condition is true. After the condition is

evaluated, a is incremented to 20 and b is decremented

to 19.

Now, printf("%d",a+++--b); is evaluated. b is decremented to 18. So, (20+18) i.e. 38 is printed. After

that, a is incremented to 21.

Hence, output is 38.

Q7 Text Solution:

```
void main()
    {
    int a=0;
    printf("%d", a); // 0 is printed
     if(a=2){//Assignment operator assigns and
returns the assigned value; So 2 is assigned to a
```

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and 2 is returned. Any non-zero value is considered true.

```
printf("Hi");//"Hi" is printed
    printf("%d",a);//Since a contains 2, 2 is printed.
}else{
    printf("Bye");
}
    printf("%d", a); //Since a contains 2, 2 is printed
}
```

Output: 0Hi22

Q8 **Text Solution:**

a = 0. b=0;

a=(a=4)||(b=1) //Assignment operator assigns and returns the assigned value. Here, short-circuiting will be applied. Since the logical operator is OR, if the first part is true, second part is not evaluated at all. Hence, b=0, a=1.

if(a && b)//The condition evaluates to 1 && 0 i.e. 0. Hence, else part is evaluated.

Output: PhysicsWalah0

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